

CSA1708-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 2 : Python Program to Solve 8-Queen Problem

Aim

To develop a Python program that places N queens on an N×N chessboard so that no two queens attack each other.

Algorithm

- Read the number of queens N from the user.
- Create a list board of size N, where board[row] stores the column of the queen in that row.
- Start from row 0 and try every column in the current row.
- A position is safe if no earlier queen is in the same column or on the same diagonal.
- If a position is safe, place the queen and move to the next row.
- If no column is safe in a row, backtrack to the previous row and try its next column.
- When all N rows have a queen, print the board with Q for queens and . for empty squares.
- If no arrangement is possible, report that no solution exists.

Program

```
n = int(input("Enter number of queens: "))

board = [-1] * n

def safe(row, col):
    return all(board[i] != col and abs(board[i] - col) != row - i
    for i in range(row))

def solve(row):
    if row == n:
        return True
    for col in range(n):
        if safe(row, col):
            board[row] = col
            if solve(row + 1):
                return True
    board[row] = -1
    return False

if solve(0):
    for col in board:
        print("." * col + "Q" + "." * (n - col - 1))
else:
    print("No solution exists")
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/shaik/OneDrive/Documents/2.py =====
Enter number of queens: 8
Q.....
..Q.....
...Q...
.Q.....
...Q...
Q.....
Q.....
Q.....
>>>
```

RESULT

Thus, the 8-Queen Problem was solved successfully using Python.